

Question Number	Answer	Additional Guidance	Mark
3(a)	one sensible risk identified	eg cuts/burns/allergies/infections/electrocution	(1) exp

Question Number	Answer	Additional Guidance	Mark
3(b)	A description that includes the following points: - find a suitable method for measuring the production of oxygen (1) - over a suitable time period (1) - find a suitable method for varying the light intensity (1)	IGNORE practise the method IGNORE rate of photosynthesis / counting oxygen bubbles ALLOW stated number of minutes / hours e.g. 10 minutes to 8 hours	(3) Exp

Question Number	Answer	Additional Guidance	Mark
3(c)	An answer that includes ten of the following points: • clear statement of the dependent variable (1) • description of method to provide 5 different intensities of light (1) • plant given time to acclimatise (1) • description of method for measuring gas volume (1) • method of calculating rate of gas released (1) • the carbon dioxide must be in excess (1) • variables that need to be taken into account (2) • description of how these variables are {controlled/measured} (2) • repeats for each light intensity (1)	ALLOW {volume / concentration} of oxygen IGNORE counting oxygen bubbles ALLOW equilibrate IGNORE counting bubbles ALLOW number of bubbles per unit time ALLOW use of sodium hydrogen carbonate e.g. temperature, pH, background light e.g. controlled room temperature or waterbath. buffer or pH meter, blackout other light sources	(10) exp

Question Number	Answer	Additional Guidance	Mark
3(d)	An answer that includes the following points: • table with headings (1) • means calculated from repeats (1) • line graph format with labelled axes (1) • use of an appropriate correlation statistical test (1)	ALLOW concentration of oxygen Units only needed once from table or graph ALLOW sketch graph either rising or falling depending on axis labels e.g. Pearsons, Spearman's Rank	(4) exp

Question Number	Answer	Additional Guidance	Mark
3(e)	An answer that includes three of the following points: • difficult to control all variables affecting {growth / photosynthesis} of plants (1) • idea of difficulty of controlling a named variable (1) • idea that surface area cannot be controlled/measured (1) • possible errors in measuring volume of gas released (1)	ALLOW biotic/ abiotic variables e.g. temperature or pH ALLOW size of leaves ALLOW size of bubble may vary Not counting bubbles unqualified	(3)

(Total for question 3 = 21 marks)

